**Education**
Computer Science and Software Engineering

**Language proficiency**
English – B2
Polish – B1
Swedish – A2

**Domains**
Telecommunications
Library Systems
UAV Systems
Aerospace Systems

**Software engineer with 7+ years of commercial experience.**
I possess a strong background in both hardware and software development. My expertise mostly revolves around C/C++ software development, where I appreciate the complexity and versatility of the language, and MATLAB, where I have extensive experience in developing algorithms and applications for image analysis. I am eager to work on projects that prioritize performance and innovative features and enthusiastic about collaborating on challenging projects that allow me to leverage the full capabilities of these technologies and contribute to cutting-edge solutions.

**Programming Languages**
C++, C, C#, Java, JavaScript, MATLAB, Scala.

**Operating Systems**
Embedded Linux.

**Tools**
Yocto Project, Qt Designer, System Tool Kit, CMake, qmake, GDB, Qt Test, Visual Studio, Qt Creator.

**Frameworks & Libraries**
GStreamer, Qt, Qt Widgets, QSS, .NET, Node.js, WinForms, V4L2.

**Multimedia**
WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient.

**Databases**
MySQL, PostgreSQL, SQLite.

**Network**
SDP, SIP, RTP, Qt WebSockets, SNMP, DHCP, TFTP.

**Cloud**
AWS.

**Embedded**
BitBake, NXP i.MX.

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**Development Processes**
Jira, Asana.

**Source control systems**
Git, GitLab, Bitbucket.

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# Projects

## SIP DESK PHONE PLATFORM FOR AUDIO AND VIDEO COMMUNICATION

A SIP desk phone platform for audio and video communication, supporting both camera-enabled and audio-only device models. The project included the native phone application, web management interface, automated provisioning tools, and embedded system customization. It covered everyday calling workflows such as SIP registration, incoming and outgoing calls, video calls, call hold, transfer, conferencing, phonebook access, direct IP dialing, remote configuration, and device setup.

**Project roles**
Software Engineer

**Period**
05.2025 – till now

**Responsibilities**

- Refactored SIP call handling toward a cleaner MVC/MVVM-style architecture by separating UI rendering from call state, user actions, media negotiation, and Linphone-SDK integration logic;
- Incrementally decomposed legacy Qt Widgets call-flow code into clearer domain, presentation, and SDK-integration boundaries while preserving existing UI behavior and QObject signal compatibility;
- Engineered real-time audio and video pipelines using GStreamer and WebRTC on an embedded Linux (NXP i.MX) architecture;
- Managed audio synchronization, jitter buffers, and media negotiation for multi-participant streams, ensuring low-latency performance on constrained hardware;
- Hardened video-call and conference UI behavior for edge cases such as held calls, second-line calls, remote hangups, stale overlays, and active video stream switching;
- Extended and customized Linphone-SDK (liblinphone) backend logic to implement SIP failover and backup-server connectivity for improved device reliability;
- Developed WebRTC video-conferencing UI and signaling flows: conference creation/join, incoming invite screens, participant roster, SIP/IP/phonebook add dialogs, status strip, control panel, video tile layout, and audio/video device controls;
- Autoprovisioning tooling: migrated a WinForms-based configuration utility from .NET Framework 4.8 to .NET 8, unified project file formats across the device ecosystem, and

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implemented provisioning workflows using SNMP, DHCP, TFTP, and SQLite;
- Collaborated on Yocto/BitBake customization, i.MX embedded Linux target integration, and device-side service configuration for hardware-software integration.

**Environment**
C++17, C#, JavaScript, Embedded Linux, Yocto, Qt Designer, CMake, qmake, gdb, QtTest, Qt Creator, GStreamer, Qt 5, Qt Widgets, QSS, .NET 8, Node.js, WinForms, V4L2, WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient, SQLite, SDP, SIP, RTP, Qt WebSockets, SNMP, DHCP, TFTP, BitBake, NXP i.MX, Asana, Git, GitLab.

## LIBRARY SYSTEM

A robust and highly scalable library management system with over 40 years of operational history, serving a diverse client base across the globe. The system is built using a combination of C, Java, and Scala, leveraging the strengths of each language to deliver a reliable, efficient, and feature-rich platform. Designed to handle high volumes of data and complex workflows, it has continually evolved to meet the needs of modern libraries while maintaining compatibility with legacy components.

**Project roles**
Software Engineer

**Period**
03.2022 – 04.2024

**Responsibilities**
- Optimized and maintained the existing codebase by resolving complex bugs and implementing performance enhancements;
- Developed new backend and frontend features for socket-based communication;
- Led development efforts as a key contributor in C language, driving the successful implementation of critical features and solutions;
- Refactored existing Scala API code to improve maintainability and scalability;
- Worked with AWS pipelines, automating CI/CD processes for smoother deployments and reducing development cycle times;
- Modified database tables to support new feature implementations, improving schema design to align with updated system workflows;
- Estimated tasks with precision, ensuring realistic timelines and efficient resource allocation for

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development milestones;
- Conducted thorough code reviews, ensuring code quality, adherence to standards, and knowledge sharing within the basic team and client's team;
- Refactored legacy application code to improve maintainability, scalability, and overall system efficiency.

**Environment**
C, Java 11, Scala, gdb, MySQL, PostgreSQL, Git, BitBucket, AWS.

## SOFTWARE MODULES FOR UAV

Developed innovative software solutions and algorithms to enhance the performance of optical sensors used in satellites and UAVs. The project included advanced photo and video processing, data analysis, and testing to improve system reliability, accuracy, and image quality for remote sensing applications.

**Project roles**
Software Engineer

**Period**
08.2021 – 12.2022

**Responsibilities**
- Developed software modules for digital image processing systems used in Unmanned Aerial Vehicles;
- Developed MATLAB scripts to process and stabilize UAV video feeds, writing algorithms to mathematically correct motion-induced visual distortions.
- Implemented a debayer algorithm to convert raw sensor data into full-color images, significantly improving image quality and accuracy for better visual data interpretation;
- Created image format conversion tools to support multiple formats, enhancing compatibility and integration across different systems and platforms;
- Developed an image stabilization algorithm to reduce motion-induced distortions, improving video clarity in real-time UAV operations for clearer footage capture;
- Designed an image fusion algorithm to combine data from multiple sensors, increasing detail and contrast for improved target recognition and analysis.

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**Environment**
MATLAB, PostgreSQL, Git, BitBucket.

## SOFTWARE FOR PROCESSING AND ANALYSIS OF SATELLITE EARTH OBSERVATION IMAGES

The software includes a range of modules with functions such as determining linear resolution, enhancing image quality through prediction of image type and other factors, stitching frames taken by a remote sensing satellite, eliminating optical distortions between frames. Additionally, one of the key components of the project is the development of a compression algorithm based on Adaptive differential pulse-code modulation (ADPCM).

These features are aimed at providing a comprehensive toolkit for working with satellite imagery, enabling precise resolution analysis, effective data compression, and optimized image processing workflows for various remote sensing applications.

**Project roles**
Software Engineer

**Period**
08.2019 – 03.2021

**Responsibilities**
- Designed C++ and MATLAB algorithms to analyze image sharpness using the sharp edge method;
- Developed algorithms for standardizing brightness levels across multiple frames and eliminating optical defects, improving the overall quality and consistency of satellite imagery;
- Designed and implemented software to analyze the linear resolution of satellite images using the sharp edge method, enhancing the accuracy of imaging performance evaluation;
- Created algorithms to assess image sharpness and edge clarity, ensuring precise resolution measurements for remote sensing applications and contributing to reliable data interpretation;
- Developed software to predict quantization ranges for remote sensing satellites by analyzing Earth observation data and estimating the albedo range, improving sensor calibration and minimizing data loss;
- Automated the prediction of quantization ranges, streamlining workflows, and improving operational efficiency;
- Designed and implemented a custom image compression algorithm using ADPCM with multiple quantizers, achieving optimized compression ratios and enhanced flexibility for different data sizes.

**Environment**
C++, MATLAB, System Tool Kit, CMake, Visual Studio, Jira, Git.

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# Professional skills

| SKILLS | | EXPERIENCE IN YEARS | LAST USED |
|---|---|---|---|
| Programming Languages | C++ | 3 | 2026 |
| | C | 2 | 2024 |
| | C# | 1 | 2026 |
| | Java | 2 | 2024 |
| | JavaScript | 1 | 2026 |
| | MATLAB | 3 | 2022 |
| | Scala | 2 | 2024 |
| Operating Systems | Embedded Linux | 1 | 2026 |
| Tools | Yocto Project | 1 | 2026 |
| | Qt Designer | 1 | 2026 |
| | System Tool Kit | 2 | 2021 |
| | CMake | 3 | 2026 |
| | qmake | 1 | 2026 |
| | GDB | 4 | 2026 |
| | Qt Test | 1 | 2026 |
| | Visual Studio | 2 | 2021 |
| | Qt Creator | 1 | 2026 |
| Frameworks & Libraries | GStreamer | 1 | 2026 |
| | Qt | 1 | 2026 |
| | Qt Widgets | 1 | 2026 |
| | QSS | 1 | 2026 |
| | .NET | 1 | 2026 |
| | Node.js | 1 | 2026 |
| | WinForms | 1 | 2026 |
| | V4L2 | 1 | 2026 |

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| SKILLS | | EXPERIENCE IN YEARS | LAST USED |
|---|---|---|---|
| Multimedia | WebRTC | 1 | 2026 |
| | Qt Multimedia | 1 | 2026 |
| | Linphone-SDK | 1 | 2026 |
| | liblinphone | 1 | 2026 |
| | mediasoup | 1 | 2026 |
| | libmediasoupclient | 1 | 2026 |
| Databases | MySQL | 2 | 2024 |
| | PostgreSQL | 3 | 2024 |
| | SQLite | 1 | 2026 |
| Network | SDP | 1 | 2026 |
| | SIP | 1 | 2026 |
| | RTP | 1 | 2026 |
| | Qt WebSockets | 1 | 2026 |
| | SNMP | 1 | 2026 |
| | DHCP | 1 | 2026 |
| | TFTP | 1 | 2026 |
| Cloud | AWS | 4 | 2024 |
| Embedded | BitBake | 1 | 2026 |
| | NXP i.MX | 1 | 2026 |
| Development Processes | Jira | 2 | 2021 |
| | Asana | 1 | 2026 |
| Source control systems | Git | 6 | 2026 |
| | GitLab | 1 | 2026 |
| | Bitbucket | 3 | 2024 |

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